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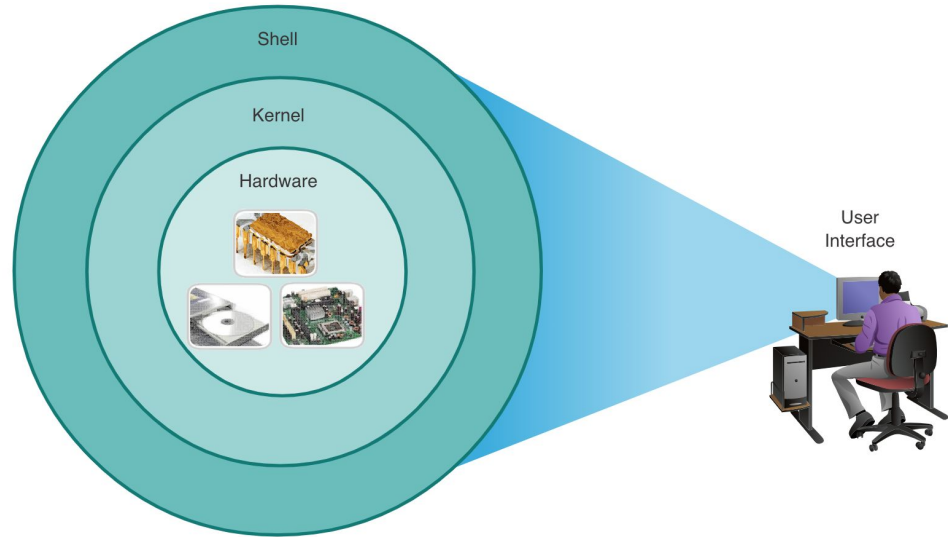
Basic End Device, Switch and Router Configuration

Cisco IOS Access: Operating Systems



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- **Shell** - The user interface that allows users to request specific tasks from the computer. These requests can be made either through the CLI or GUI interfaces.
- **Kernel** - Communicates between the hardware and software of a computer and manages how hardware resources are used to meet software requirements.
- **Hardware** - The physical part of a computer including underlying electronics.



Cisco IOS Access: GUI



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- A GUI allows the user to interact with the system using an environment of graphical icons, menus, and windows.
- A GUI is more user-friendly and requires less knowledge of the underlying command structure that controls the system.
- Examples of these are: Windows, macOS, Linux KDE, Apple iOS and Android.
- GUIs can fail, crash, or simply not operate as specified. For these reasons, network devices are typically accessed through a CLI.



Cisco IOS Access: Purpose of an OS



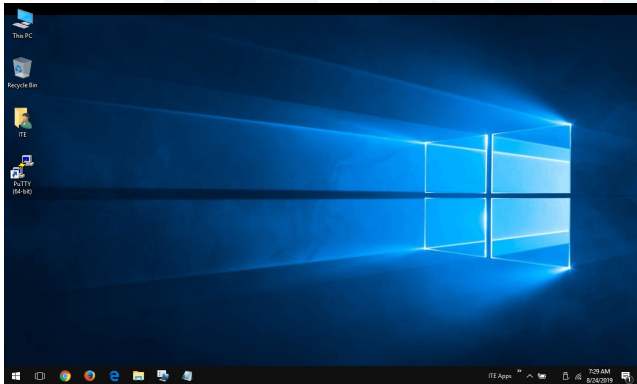
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PC operating system enables a user to do the following:

- Use a mouse to make selections and run programs
- Enter text and text-based commands
- View output on a monitor

CLI-based network operating system enables a network technician to do the following:

- Use a keyboard to run CLI-based network programs
- Use a keyboard to enter text and text-based commands
- View output on a monitor



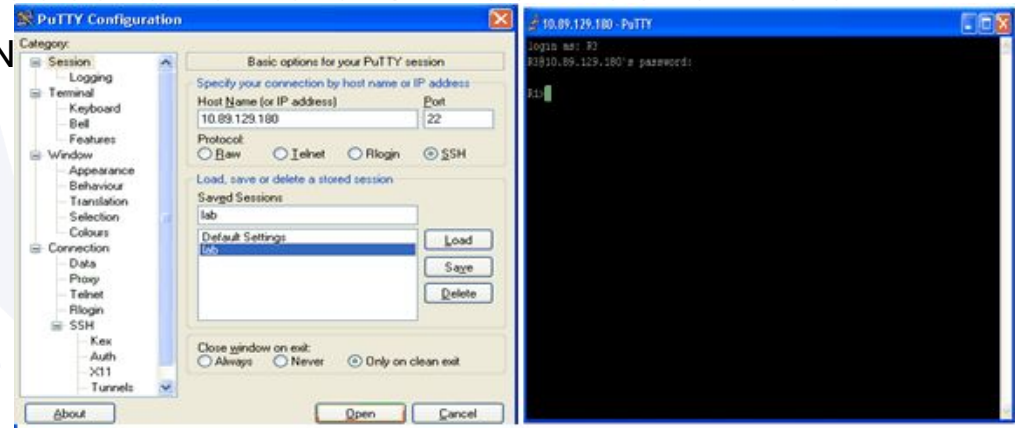
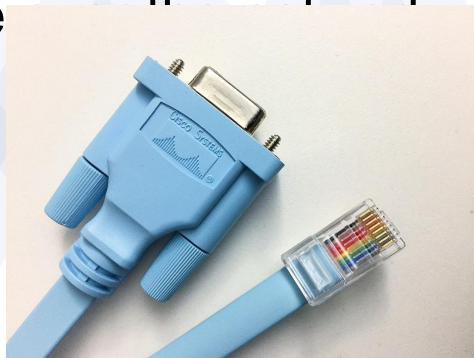
```
analyst@secOps ~]$ ls
Desktop Downloads lab.support.files second_drive
[analyst@secOps ~]$
```

Cisco IOS Access: Access Methods



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- **Console** – A physical management port used to access a device in order to provide maintenance, such as performing the initial configurations.
- **Secure Shell (SSH)** – Establishes a secure remote CLI connection to a device, through a virtual interface, over a network.
(Note: This is the recommended method for remotely connecting to a device.)
- **Telnet** – Establishes an insecure remote CLI connection to a device sent over

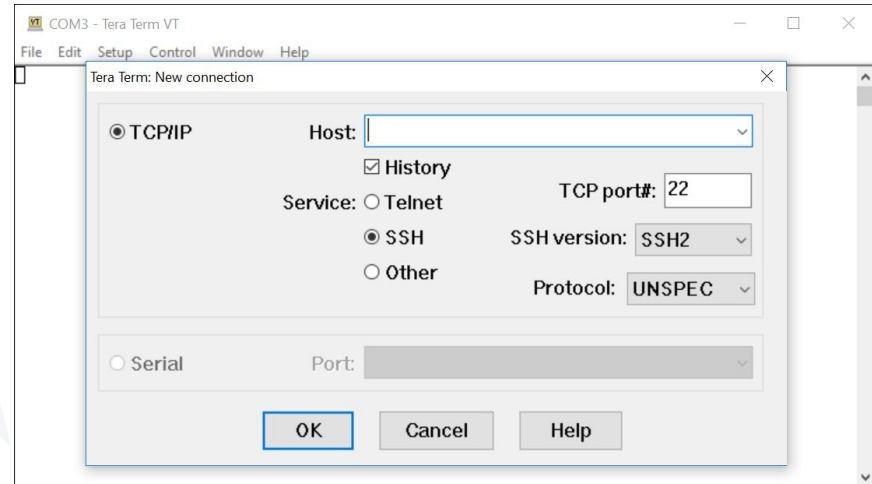
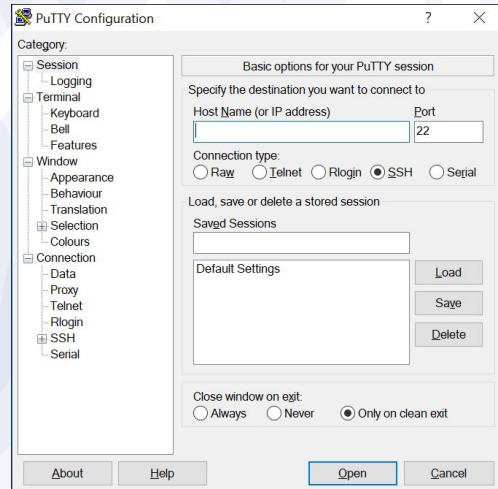


Cisco IOS Access: Terminal Emulation Programs



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- Terminal emulation programs are used to connect to a network device by either a console port or by an SSH/Telnet connection.
- There are several terminal emulation programs to choose from such as PuTTY, Tera Term and SecureCRT.





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IOS Navigation

IOS Navigation: Primary Command Modes



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User EXEC Mode:

- Allows access to only a limited number of basic monitoring commands
- Identified by the CLI prompt that ends with the > symbol

```
Router>
```

```
Switch>
```

Privileged EXEC Mode:

- Allows access to all commands and features
- Identified by the CLI prompt that ends with the # symbol

```
Router#
```

```
Switch#
```

IOS Navigation: Configuration Mode and Subconfiguration Modes



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Global Configuration Mode:

- Used to access configuration options on the device

```
Switch(config) #
```

Line Configuration Mode:

- Used to configure console, SSH, Telnet or AUX access

```
Switch(config-line) #
```

Interface Configuration Mode:

- Used to configure a switch port or router interface

```
Switch(config-if) #
```

IOS Navigation: Navigation Between IOS Modes



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- **Privileged EXEC Mode:**
 - To move from user EXEC mode to privilege EXEC mode, use the **enable** command.
- **Global Configuration Mode:**
 - To move in and out of global configuration mode, use the **configure terminal** command. To return to privilege EXEC mode, use the **exit** command.
- **Line Configuration Mode:**
 - To move in and out of line configuration mode, use the **line** command followed by the management line type. To return to global configuration mode, use the **exit** command.

```
Switch> enable  
Switch#
```

```
Switch(config)#  
Switch(config)#exit  
Switch#
```

```
Switch(config)#line console 0  
Switch(config-line)#exit  
Switch(config)#
```

IOS Navigation: Navigation Between IOS Modes (Cont.)



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Subconfiguration Modes:

- To move out of any subconfiguration mode to get back to global configuration mode, use the **exit** command. To return to privilege EXEC mode, use the **end** command or key combination **Ctrl +Z**.

```
Switch(config)#line console 0
Switch(config-line)#end
Switch#
```

- To move directly from one subconfiguration mode to another, type in the desired subconfiguration mode command. In the example, the command prompt changes from **(config-line)#** to **(config-if)#**.

```
Switch(config-line)#interface FastEthernet 0/1
Switch(config-if)#
```



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The Command Structure

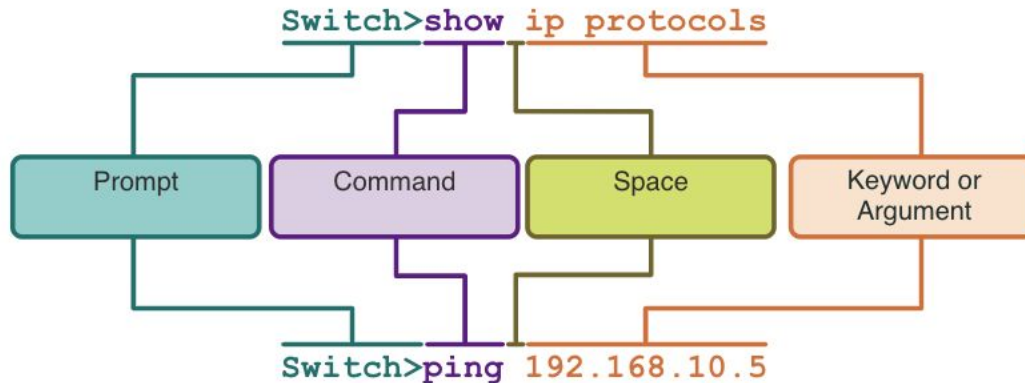
The Command Structure: Basic IOS

Command Structure



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- **Keyword** – This is a specific parameter defined in the operating system (in the figure, **ip protocols**).
- **Argument** - This is not predefined; it is a value or variable defined by the user (in the figure, **192.168.10.5**).



The Command Structure: IOS Command Syntax Check



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A command might require one or more arguments. To determine the keywords and arguments required for a command, refer to the command syntax.

- Boldface text indicates commands and keywords that are entered as shown.
- Italic text indicates an argument for which the user provides the value.

Convention	Description
boldface	Boldface text indicates commands and keywords that you enter literally as shown.
<i>italics</i>	Italic text indicates arguments for which you supply values.
[x]	Square brackets indicate an optional element (keyword or argument).
{x}	Braces indicate a required element (keyword or argument).
[x {y z }]	Braces and vertical lines within square brackets indicate a required choice within an optional element. Spaces are used to clearly delineate parts of the command.

The Command Structure: IOS Command Syntax Check (Cont.)



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- The command syntax provides the pattern, or format, that must be used when entering a command.
- The command is **ping** and the user-defined argument is the *ip-address* of the destination device. For example, **ping 10.10.10.5**.
- The command is **traceroute** and the user-defined argument is the *ip-address* of the destination device. For example, **traceroute 192.168.254.254**.
- If a command is complex with multiple arguments, you may see it represented like this:

```
ping ip-address
```

```
traceroute ip-address
```

```
Switch(config-if)# switchport port-security aging { static | time time | type {absolute | inactivity}}
```

The Command Structure: IOS Help Features



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The IOS has two forms of help available: context-sensitive help and command syntax check.

- Context-sensitive help enables you to quickly find answers to these questions:
 - Which commands are available in each command mode?
 - Which commands start with specific characters or group of characters?
 - Which arguments and keywords are available to particular commands?
- Command syntax check verifies that a valid command was entered by the user.
 - If the interpreter cannot understand the command being entered, it will provide feedback describing what is wrong with the command.

```
Router#ping ?  
WORD  Ping destination address or hostname  
ip     IP echo  
ipv6   IPv6 echo
```

```
Switch#interface fastEthernet 0/1  
      ^  
% Invalid input detected at '^' marker.
```

The Command Structure: Hot Keys and Shortcuts



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- The IOS CLI provides hot keys and shortcuts that make configuring, monitoring, and troubleshooting easier.
- Commands and keywords can be shortened to the minimum number of characters that identify a unique selection. For example, the **configure** command can be shortened to **conf** because **configure** is the only command that begins with **conf**.

```
Router#con
% Ambiguous command: "con"
Router#con?
configure  connect
```

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
```

The Command Structure: Hot Keys and Shortcuts (Cont.)



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- The table below is a brief list of keystrokes to enhance command line editing.

Keystroke	Description
Tab	Completes a partial command name entry.
Backspace	Erases the character to the left of the cursor.
Left Arrow or Ctrl+B	Moves the cursor one character to the left.
Right Arrow or Ctrl+F	Moves the cursor one character to the right.
Up Arrow or Ctrl+P	Recalls the commands in the history buffer, beginning with the most recent commands.

The Command Structure: Hot Keys and Shortcuts (Cont.)



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- When a command output produces more text than can be displayed in a terminal window, the IOS will display a “**--More--**” prompt.

The table below describes the keystrokes that can be used when this prompt is displayed.

Keystroke	Description
Enter Key	Displays the next line.
Space Bar	Displays the next screen.
Any other key	Ends the display string, returning to privileged EXEC mode.

- The table below lists commands that can be used to exit out of an operation.

Keystroke	Description
Ctrl-C	When in any configuration mode, ends the configuration mode and returns to privileged EXEC mode.
Ctrl-Z	When in any configuration mode, ends the configuration mode and returns to privileged EXEC mode.
Ctrl-Shift-6	All-purpose break sequence used to abort DNS lookups, traceroutes, pings, etc.



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Basic Device Configuration

Basic Device Configuration

Device Names



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- The first configuration command on any device should be to give it a unique hostname.
- By default, all devices are assigned a factory default name. For example, a Cisco IOS switch is "Switch."
- Guideline for naming devices:
 - Start with a letter
 - Contain no spaces
 - End with a letter or digit
 - Use only letters, digits, and dashes
 - Be less than 64 characters in length

```
Switch# configure terminal
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
```

Note: To return the switch to the default prompt, use the **no hostname** global config command.

Basic Device Configuration

Password Guidelines



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- The use of weak or easily guessed passwords are a security concern.
 - All networking devices should limit administrative access by securing privileged EXEC, user EXEC, and remote Telnet access with passwords. In addition, all passwords should be encrypted and legal notifications provided.
-
- Password Guidelines:
 - Use passwords that are more than eight characters in length.
 - Use a combination of upper and lowercase letters, numbers, special characters, and/or numeric sequences.
 - Avoid using the same password for all devices.
 - Do not use common words because they are easily guessed.



Note: Most of the labs in this course use simple passwords such as **cisco** or **class**. These passwords are considered weak and easily guessable and should be avoided in production environments.

Basic Device Configuration

Configure Passwords



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Securing user EXEC mode access:

- First enter line console configuration mode using the **line console 0** command in global configuration mode.
- Next, specify the user EXEC mode password using the **password** *password* command.
- Finally, enable user EXEC access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line console 0
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Securing privileged EXEC mode access:

- First enter global configuration mode.
- Next, use the **enable secret** *password* command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# enable secret class
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

Basic Device Configuration

Configure Passwords (Cont.)



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Securing VTY line access:

- First enter line VTY configuration mode using the **line vty 0 15** command in global configuration mode.
- Next, specify the VTY password using the **password *password*** command.
- Finally, enable VTY access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line vty 0 15
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

- Note: VTY lines enable remote access using Telnet or SSH to the device. Many Cisco switches support up to 16 VTY lines that are numbered 0 to 15.

Basic Device Configuration

Encrypt Passwords



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- The startup-config and running-config files display most passwords in plaintext.
 - To encrypt all plaintext passwords, use the **service password-encryption** global config command.
-
- Use the **show running-config** command to verify that the passwords on the device are now encrypted.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# service password-encryption
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

```
Sw-Floor-1# show running-config
!
!
line con 0
password 7 094F471A1A0A
login
!
Line vty 0 4
Password 7 03095A0F034F38435B49150A1819
Login
!
!
end
```

Basic Device Configuration

Banner Messages



- A banner message is important to warn unauthorized personnel from attempting to access the device.
- To create a banner message of the day on a network device, use the **banner motd # the message of the day #** global config command.

The banner will be displayed on attempts to access the device.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# banner motd #Authorized Access Only!#
```



```
Press RETURN to get started.
```

```
Authorized Access Only!
```

```
User Access Verification
```

```
Password:
```

Note: The “#” in the command syntax is called the delimiting character. It is entered before and after the message.



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Save Configurations

Save Configurations: Configuration Files



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- There are two system files that store the device configuration:
 - **startup-config** - This is the saved configuration file that is stored in NVRAM. It contains all the commands that will be used by the device upon startup or reboot. Flash does not lose its contents when the device is powered off.
 - **running-config** - This is stored in Random Access Memory (RAM). It reflects the current configuration. Modifying a running configuration affects the operation of a Cisco device immediately. RAM is volatile memory. It loses all of its content when the device is powered off or restarted.
 - To save changes made to the running configuration to the startup configuration file, use the `copy running-config startup-config` privileged EXEC command.

```
Router#show startup-config
Using 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
Router#show running-config
Building configuration...

Current configuration : 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

Save Configurations: Alter the Running Configurations



If changes made to the running config do not have the desired effect and the running-config has not yet been saved, you can restore the device to its previous configuration. To do this you can:

- Remove the changed commands individually.
- Reload the device using the **reload** command in privilege EXEC mode.
Note: This will cause the device to briefly go offline, leading to network downtime.

If the undesired changes were saved to the startup-config, it may be necessary to clear all the configurations using the **erase startup-config** command in privilege EXEC mode.

- After erasing the startup-config, reload the device to clear the running-config file from RAM.

```
Router# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Router#
```

```
Router# reload
Proceed with reload? [confirm]
Initializing Hardware ...
```

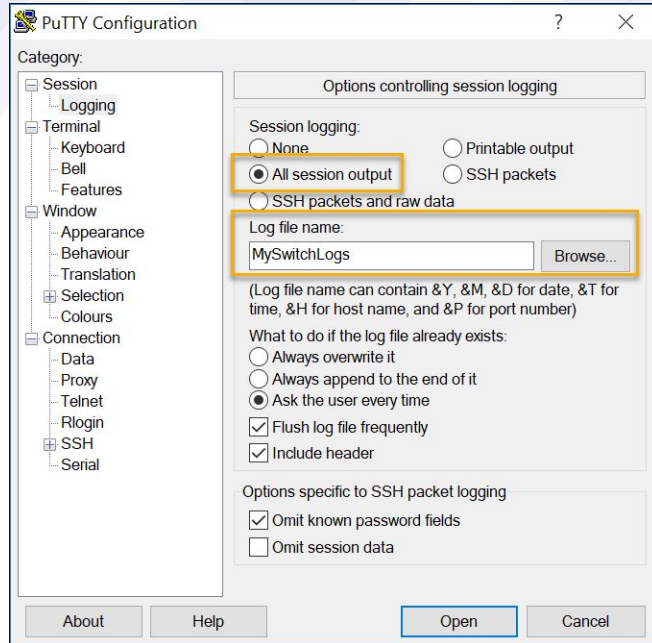
Save Configurations: Capture Configuration to a Text File



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Configuration files can also be saved and archived to a text document.

- **Step 1.** Open terminal emulation software, such as PuTTY or Tera Term, that is already connected to a switch.
- **Step 2.** Enable logging in to the terminal software and assign a name and file location to save the log file. The figure displays that **All session output** will be captured to the file specified (i.e., MySwitchLogs).

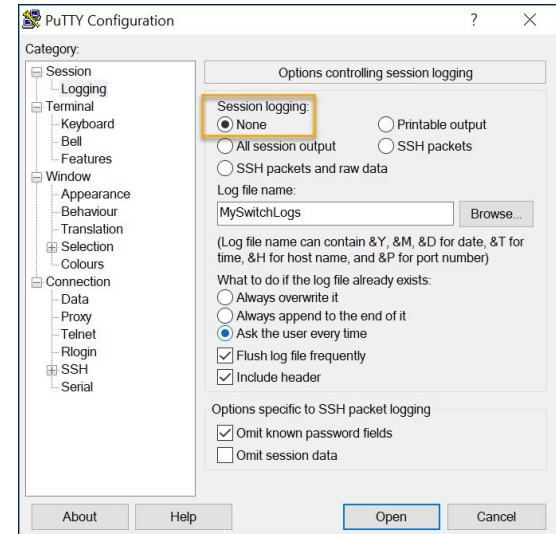




- **Step 3.** Execute the **show running-config** or **show startup-config** command at the privileged EXEC prompt. Text displayed in the terminal window will be saved to the text file.
- **Step 4.** Disable logging in the terminal software. The figure shows how to disable logging by choosing the **None** session logging option

```
Switch# show running-config  
Building configuration...
```

Note: The text file created can be used as a record of how the device is currently implemented. The file could require editing before being used to restore a saved configuration to a device.





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Ports and Addresses

Ports and Addresses

IP Addresses



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- The use of IP addresses is the primary means of enabling devices to locate one another and establish end-to-end communication on the internet.
- The structure of an IPv4 address is called dotted decimal notation and is represented by four decimal numbers between 0 and 255.
- An IPv4 subnet mask is a 32-bit value that differentiates the network portion of the address from the host portion. Coupled with the IPv4 address, the subnet mask determines to which subnet the device is a member.
- The default gateway address is the IP address of the router that the host will use to access remote networks, including the internet.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 10

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

Ports and Addresses

IP Addresses (Cont.)



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- IPv6 addresses are 128 bits in length and written as a string of hexadecimal values. Every four bits is represented by a single hexadecimal digit; for a total of 32 hexadecimal values. Groups of four hexadecimal digits are separated by a colon ":".
- IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.

Note: IP in this course refers to both the IPv4 and IPv6 protocols. IPv6 is the most recent version of IP and is replacing the more common IPv4.

Internet Protocol Version 6 (TCP/IPv6) Properties

General

You can get IPv6 settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IPv6 settings.

☐ Obtain an IPv6 address automatically

☒ Use the following IPv6 address:

IPv6 address: 2001:db8:acad:10::10

Subnet prefix length: 64

Default gateway: fe80::1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

☐ Validate settings upon exit

Advanced...

OK Cancel

Ports and Addresses

Interfaces and Ports

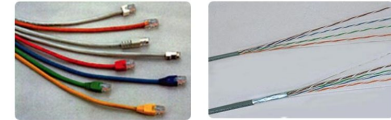


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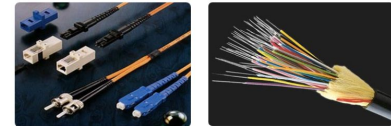
- Network communications depend on end user device interfaces, networking device interfaces, and the cables that connect them.
- Types of network media include twisted-pair copper cables, fiber-optic cables, coaxial cables, or wireless.
- Different types of network media have different features and benefits. Some of the differences between various types of media include:
 - ❖ Distance the media can successfully carry a signal
 - ❖ Environment in which the media is to be installed
 - ❖ Amount of data and the speed at which it must be transmitted
 - ❖ Cost of the media and installation



Copper



Fiber-optics



Wireless





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Configure IP Addressing

Configure IP Addressing

Manual IP Address Configuration for End Devices



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- End devices on the network need an IP address in order to communicate with other devices on the network.
- IPv4 address information can be entered into end devices manually, or automatically using Dynamic Host Configuration Protocol (DHCP).
 - To manually configure an IPv4 address on a Windows PC, open the **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next right-click and select **Properties** to display the **Local Area Connection Properties**.
 - Next, click **Properties** to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** window. Then configure the IPv4 address and subnet mask information, and default gateway.

Note: IPv6 addressing, and configuration options are similar to

Configure IP Addressing

Automatic IP Address Configuration for End Devices



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- DHCP enables automatic IPv4 address configuration for every end device that is DHCP-enabled.
- End devices are typically by default using DHCP for automatic IPv4 address configuration.
 - To configure DHCP on a Windows PC, open the **Control Panel > Network Sharing Center > Change adapter settings** and choose the adapter. Next right-click and select **Properties** to display the **Local Area Connection Properties**.
 - Next, click **Properties** to open the **Internet Protocol Version 4 (TCP/IPv4) Properties** window, then select **Obtain an IP address automatically** and **Obtain DNS server address automatically**.



Note: IPv6 uses DHCPv6 and SLAAC (Stateless Address Autoconfiguration) for dynamic address allocation.

Configure IP Addressing

Switch Virtual Interface Configuration



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To access the switch remotely, an IP address and a subnet mask must be configured on the SVI.

To configure an SVI on a switch:

- Enter the **interface vlan 1** command in global configuration mode.
- Next assign an IPv4 address using the **ip address** *ip-address subnet-mask command*.
- Finally, enable the virtual interface using the **no shutdown** command.

```
Switch# configure terminal
Switch(config)# interface vlan 1
Switch(config-if)# ip address 192.168.1.20 255.255.255.0
Switch(config-if)# no shutdown
```



Router Configs

Basic Router Configuration Steps



- Configure the device name.
- Secure privileged EXEC mode.
- Secure user EXEC mode.
- Secure remote Telnet / SSH access.
- Encrypt all plaintext passwords.
- Provide legal notification and save the

```
Router(config)# hostname hostname
```

```
Router(config)# enable secret password
```

```
Router(config)# line console 0  
Router(config-line)# password password  
Router(config-line)# login
```

```
Router(config)# line vty 0 4  
Router(config-line)# password password  
Router(config-line)# login  
Router(config-line)# transport input {ssh | telnet}
```

```
Router(config)# service password encryption
```

```
Router(config)# banner motd # message #  
Router(config)# end  
Router# copy running-config startup-config
```

Basic Router Configuration Example



- Commands for basic router configuration on R1.
- Configuration is saved to NVRAM.

```
R1(config)# hostname R1
R1(config)# enable secret class
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# transport input ssh telnet
R1(config-line)# exit
R1(config)# service password encryption
R1(config)# banner motd #
Enter TEXT message. End with a new line and the #
*****
WARNING: Unauthorized access is prohibited!
*****
R1(config)# exit
R1# copy running-config startup-config
```



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Configure Interfaces

Configure Router Interfaces



- Routers support LANs and WANs and can interconnect different types of networks; therefore, they support many types of interfaces. For example, G2 ISRs have one or two integrated Gigabit Ethernet interfaces and High-Speed WAN Interface Card (HWIC) slots to accommodate other types of network interfaces, including serial, DSL, and cable interfaces.
- To be available, an interface must be:
 - **Configured with at least one IP address** - Use the **ip address** *ip-address subnet-mask* and the **ipv6 address** *ipv6-address/prefix* interface configuration commands.
 - **Activated** - By default, LAN and WAN interfaces are not activated (**shutdown**). To enable an interface, it must be activated using the **no shutdown** command. (This is similar to powering on the interface.) The interface must also be connected to another device (a hub, a switch, or another router) for the physical layer to be active.
 - **Description** - Optionally, the interface could also be configured with a short description of up to 240 characters. It is good practice to configure a description on each interface. On production networks, the benefits of interface descriptions are quickly realized as they are helpful in troubleshooting and in identifying a third-party connection and contact information.

Configure Router Interfaces

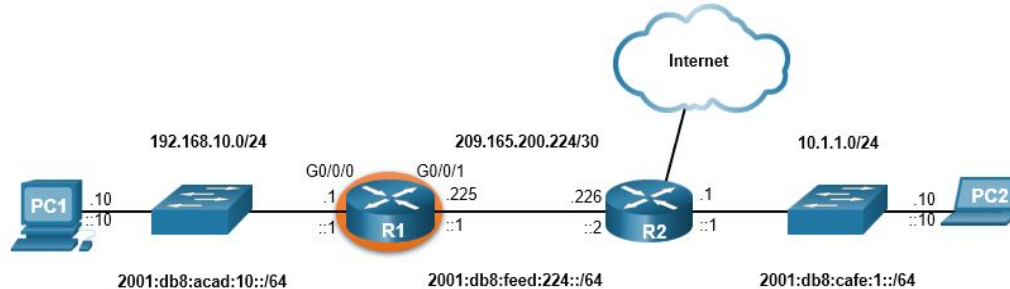


- Configuring a router interface includes issuing the following commands:

```
Router(config)# interface type-and-number  
Router(config-if)# description description-text  
Router(config-if)# ip address ipv4-address subnet-mask  
Router(config-if)# ipv6 address ipv6-address/prefix-length  
Router(config-if)# no shutdown
```

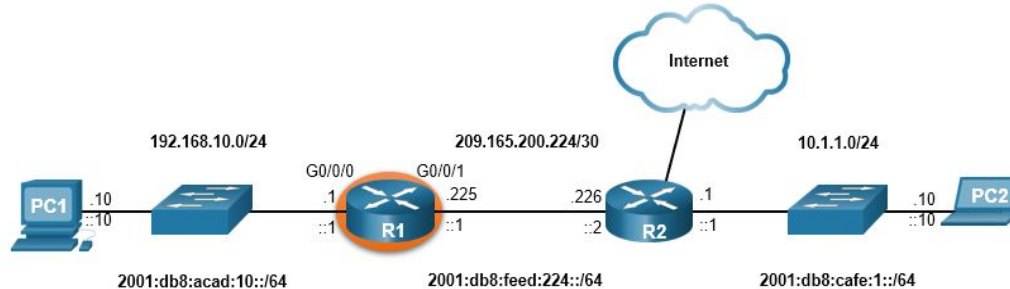
- It is a good practice to use the **description** command to add information about the network connected to the interface.
- The **no shutdown** command activates the interface.

- The commands to configure interface G0/0/0 on R1 are shown here:



```
R1(config)# interface gigabitEthernet 0/0/0
R1(config-if)# description Link to LAN
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:10::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug  1 01:43:53.435: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down
*Aug  1 01:43:56.447: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
*Aug  1 01:43:57.447: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
```

- The commands to configure interface G0/0/1 on R1 are shown here:



```
R1(config)# interface gigabitEthernet 0/0/1
R1(config-if)# description Link to R2
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:feed:224::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
*Aug 1 01:46:29.170: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
*Aug 1 01:46:32.171: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
*Aug 1 01:46:33.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1,
changed state to up
```

Configure Router Interfaces (Cont.)



- The example shows the configure for the interfaces on R1:

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# description Link to LAN 1
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ip address 192.168.11.1 255.255.255.0
R1(config-if)# ipv6 address 2001:db8:acad:2::1/64
R1(config-if)# description Link to LAN 2
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/0/0
R1(config-if)# ip address 209.165.200.225 255.255.255.252
R1(config-if)# ipv6 address 2001:db8:acad:3::225/64
R1(config-if)# description Link to R2
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)#
```

Basic Router Configuration

IPv4 Loopback Interfaces



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Another common configuration of Cisco IOS routers is enabling a loopback interface.

- The loopback interface is a logical interface that is internal to the router. It is not assigned to a physical port and can never be connected to any other device. It is considered a software interface that is automatically placed in an “up” state, as long as the router is functioning.
- The loopback interface is useful in testing and managing a Cisco IOS device because it ensures that at least one interface will always be available. For example, it can be used for testing purposes, such as testing internal routing processes, by emulating networks behind the router.
- Loopback interfaces are also commonly used in lab environments to create additional interfaces. For example, you can create multiple loopback interfaces on a router to simulate more networks for configuration practice and testing purposes. The IPv4 address for each loopback interface must be unique and unused by any other interface. In this curriculum, we often use a loopback interface to simulate a link to the internet.
- Enabling and assigning a loopback address is simple:

```
Router(config)# interface loopback number
```

```
Router(config-if)# ip address ip-address subnet-mask
```

Interface Verification Commands



- There are several **show** commands that can be used to verify the operation and configuration of an interface.
- The following commands are especially useful to quickly identify the status of an interface:
 - **show ip interface brief** and **show ipv6 interface brief** - These display a summary for all interfaces including the IPv4 or IPv6 address of the interface and current operational status.
 - **show running-config interface *interface-id*** - This displays the commands applied to the specified interface.
 - **show ip route** and **show ipv6 route** - These display the contents of the IPv4 or IPv6 routing table stored in RAM. In Cisco IOS 15, active interfaces should appear in the routing table with two related entries identified by the code '**C**' (Connected) or '**L**' (Local). In previous IOS versions, only a single entry with the code '**C**' will appear.

Verify Interface Configuration



- To verify interface configuration, use the **show ip interface brief** and **show ipv6 interface brief** commands shown here:

```
R1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0/1	209.165.200.225	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
R1# show ipv6 interface brief
```

```
GigabitEthernet0/0/0    [up/up]  
    FE80::201:C9FF:FE89:4501  
    2001:DB8:ACAD:10::1  
GigabitEthernet0/0/1    [up/up]  
    FE80::201:C9FF:FE89:4502  
    2001:DB8:FEED:224::1  
Vlan1                   [administratively down/down]  
    unassigned  
R1#
```



The table summarizes show commands used to verify interface configuration.

Commands	Description
<code>show ip interface brief</code> <code>show ipv6 interface brief</code>	Displays all interfaces, their IP addresses, and their current status.
<code>show ip route</code> <code>show ipv6 route</code>	Displays the contents of the IP routing tables stored in RAM.
<code>show interfaces</code>	Displays statistics for all interfaces on the device. Only displays the IPv4 addressing information.
<code>show ip interfaces</code>	Displays the IPv4 statistics for all interfaces on a router.
<code>show ipv6 interfaces</code>	Displays the IPv6 statistics for all interfaces on a router.

Configure Verification Commands (Cont.)



View status of all interfaces with the **show ip interface brief** and **show ipv6 interface brief** commands, shown here:

```
R1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0/1	209.165.200.225	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
R1#
```

```
R1# show ipv6 interface brief
```

```
GigabitEthernet0/0/0    [up/up]
    FE80::201:C9FF:FE89:4501
    2001:DB8:ACAD:10::1
GigabitEthernet0/0/1    [up/up]
    FE80::201:C9FF:FE89:4502
    2001:DB8:FEED:224::1
Vlan1                   [administratively down/down]
    unassigned
R1#
```

Configure Verification Commands (Cont.)



Display the contents of the IP routing tables with the **show ip route** and **show ipv6 route** commands as shown here:

```
R1# show ip route
< output omitted >
Gateway of last resort is not set
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
        209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/30 is directly connected, GigabitEthernet0/0/1
L       209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
R1#
```

```
R1# show ipv6 route
<output omitted>
C   2001:DB8:ACAD:10::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:10::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
C   2001:DB8:FEED:224::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L   2001:DB8:FEED:224::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
L   FF00::/8 [0/0]
    via Null0, receive
R1#
```

Configure Interfaces

Configure Verification Commands (Cont.)



Display statistics for all interfaces with the **show interfaces** command, as shown here:

```
R1# show interfaces gig0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Hardware is ISR4321-2x1GE, address is a0e0.af0d.e140 (bia a0e0.af0d.e140)
  Description: Link to LAN
  Internet address is 192.168.10.1/24
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive not supported
  Full Duplex, 100Mbps, link type is auto, media type is RJ45
  output flow-control is off, input flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:01, output 00:00:35, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output      drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    1180 packets input, 109486 bytes, 0 no buffer
    Received 84 broadcasts (0 IP multicasts)
    0 runts, 0 giants, 0 throttles

<output omitted>

R1#
```

Configure Interfaces (Cont.)

Display IPv4 statistics for router interfaces with the **show ip interface** command, as shown here:



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```
R1# show ip interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing Common access list is not set
  Outgoing access list is not set
  Inbound Common access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP Flow switching is disabled
```

<output omitted>

R1#

Configure Interfaces (Cont.)

Configure Verification Commands (Cont.)



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Display IPv6 statistics for router interfaces with the **show ipv6 interface** command shown here:

```
R1# show ipv6 interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is
FE80::868A:8DFF:FE44:49B0
  No Virtual link-local address(es):
  Description: Link to LAN
  Global unicast address(es):
    2001:DB8:ACAD:10::1, subnet is 2001:DB8:ACAD:10::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
    FF02::1:FF44:49B0
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND NS retransmit interval is 1000 milliseconds

R1#
```



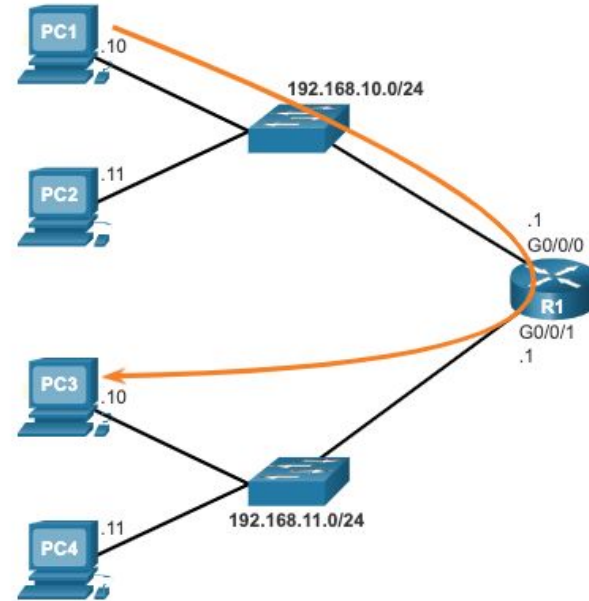
Configure the Default Gateway

Configure the Default Gateway

Default Gateway on a Host



- The default gateway is used when a host sends a packet to a device on another network.
- The default gateway address is generally the router interface address attached to the local network of the host.
- To reach PC3, PC1 addresses a packet with the IPv4 address of PC3, but forwards the packet to its default gateway, the G0/0/0 interface of R1.



Note: The IP address of the host and the router interface must be in the same network.

Configure the Default Gateway

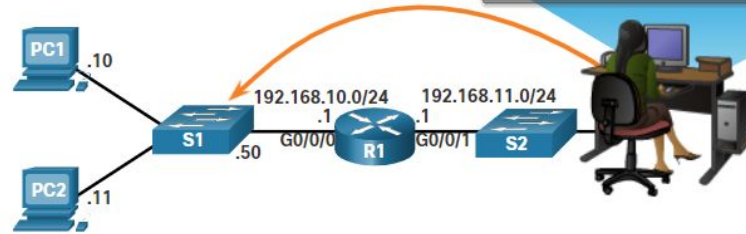
Default Gateway on a Switch



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- A switch must have a default gateway address configured to remotely manage the switch from another network.
- To configure an IPv4 default gateway on a switch, use the **ip default-gateway** *ip-address* global configuration command.

```
S1# show running-config
Building configuration...
!
<Output Omitted>
service password-encryption
!
hostname S1
!
interface Vlan1
  ip address 192.168.10.50.255.255.0
!
<Output Omitted>
!
ip default-gateway 192.168.10.1
<Output Omitted>
```



Verify Directly Connected Networks

Verify Interface Status



- The output of the **show ip interface brief** and **show ipv6 interface brief** commands can be used to quickly reveal the status of all interfaces on the router. You can verify that the interfaces are active and operational as indicated by the Status of “up” and Protocol of “up”, as shown in the example. A different output would indicate a problem with either the configuration or the cabling.

```
R1# show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0/0     192.168.10.1    YES manual up              up
GigabitEthernet0/0/1     192.168.11.1    YES manual up              up
Serial0/1/0              209.165.200.225 YES manual up              up
Serial0/1/1              unassigned      YES unset  administratively down down
R1# show ipv6 interface brief
GigabitEthernet0/0/0     [up/up]
FE80::7279:B3FF:FE92:3130
2001:DB8:ACAD:1::1
GigabitEthernet0/0/1     [up/up]
FE80::7279:B3FF:FE92:3131
2001:DB8:ACAD:2::1
Serial0/1/0              [up/up]
FE80::7279:B3FF:FE92:3130
2001:DB8:ACAD:3::1
Serial0/1/1              [down/down]    Unassigned
```

Verify IPv6 Link Local and Multicast Addresses

- The output of the **show ipv6 interface brief** command displays two configured IPv6 addresses per interface. One address is the IPv6 global unicast address that was manually entered. The other address, which begins with FE80, is the link-local unicast address for the interface. A link-local address is automatically added to an interface whenever a global unicast address is assigned. An IPv6 network interface is required to have a link-local address, but not necessarily a global unicast address.
- The **show ipv6 interface gigabitethernet 0/0/0** command displays the interface status and all of the IPv6 addresses belonging to the interface. Along with the link local address and global unicast address, the output includes the multicast addresses assigned to the interface, beginning with prefix FF02, as shown in the example.

```
R1# show ipv6 interface gigabitethernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
IPv6 is enabled, link-local address is FE80::7279:B3FF:FE92:3130
No Virtual link-local address(es):
Global unicast address(es):
  2001:DB8:ACAD:1::1, subnet is 2001:DB8:ACAD:1::/64
Joined group address(es):
  FF02::1
  FF02::1:FF00:1
  FF02::1:FF92:3130
MTU is 1500 bytes
ICMP error messages limited to one every 100 milliseconds
```

Verify Interface Configuration



- The output of the **show running-config interface** command displays the current commands applied to the specified interface, as shown.
- The following two commands are used to gather more detailed interface information:
 - **show interfaces**- Displays interface information and packet flow count for all interfaces on the device.
 - **show ip interface** and **show ipv6 interface** - Displays the IPv4 and IPv6 related information for all interfaces on a router..

```
R1 show running-config interface gigabitethernet 0/0/0
Building configuration...
Current configuration : 158 bytes
!
interface GigabitEthernet0/0/0
  description Link to LAN 1
  ip address 192.168.10.1 255.255.255.0
  negotiation auto
  ipv6 address 2001:DB8:ACAD:1::1/64
end
R1#
```

Verify Directly Connected Networks

Verify Routes



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- The output of the **show ip route** and **show ipv6 route** commands reveal the three directly connected network entries and the three local host route interface entries, as shown in the example.
- The local host route has an administrative distance of 0. It also has a /32 mask for IPv4, and a /128 mask for IPv6. The local host route is for routes on the router that owns the IP address. It is used to allow the router to process packets destined to that IP.

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

Gateway of last resort is not set

  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
  192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.11.1/32 is directly connected, GigabitEthernet0/0/1
  209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/30 is directly connected, Serial0/1/0
L       209.165.200.225/32 is directly connected, Serial0/1/0A
R1# show ipv6 route
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

C   2001:DB8:ACAD:1::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:1::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
C   2001:DB8:ACAD:2::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:2::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:3::/64 [0/0]
    via Serial0/1/0, directly connected
L   2001:DB8:ACAD:3::1/128 [0/0]
    via Serial0/1/0, receive
L   FF00::/8 [0/0]
    via Null0, receive
R1#
```

Verify Directly Connected Networks

Verify Routes (Cont.)



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- A 'C' next to a route within the routing table indicates that this is a directly connected network. When the router interface is configured with a global unicast address and is in the "up/up" state, the IPv6 prefix and prefix length are added to the IPv6 routing table as a connected route.
- The IPv6 global unicast address applied to the interface is also installed in the routing table as a local route. The local route has a /128 prefix. Local routes are used by the routing table to efficiently process packets with the interface address of the router as the destination.

```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

Gateway of last resort is not set
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
  192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.11.1/32 is directly connected, GigabitEthernet0/0/1
  209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/30 is directly connected, Serial0/1/0
L       209.165.200.225/32 is directly connected, Serial0/1/0A

R1# show ipv6 route
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

C       2001:DB8:ACAD:1::/64 [0/0]
        via GigabitEthernet0/0/0, directly connected
L       2001:DB8:ACAD:1::1/128 [0/0]
        via GigabitEthernet0/0/0, receive
C       2001:DB8:ACAD:2::/64 [0/0]
        via GigabitEthernet0/0/1, directly connected
L       2001:DB8:ACAD:2::1/128 [0/0]
        via GigabitEthernet0/0/1, receive
C       2001:DB8:ACAD:3::/64 [0/0]
        via Serial0/1/0, directly connected
L       2001:DB8:ACAD:3::1/128 [0/0]
        via Serial0/1/0, receive
L       FF00::/8 [0/0]
        via Null0, receive

R1#
```



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Filter Show Command Output



- Commands that generate multiple screens of output are, by default, paused after 24 lines. At the end of the paused output, the --More-- text displays. Pressing **Enter** displays the next line and pressing the spacebar displays the next set of lines. Use the **terminal length** command to specify the number of lines to be displayed. A value of 0 (zero) prevents the router from pausing between screens of output.
- Another very useful feature that improves the user experience in the CLI is the filtering of **show** output. Filtering commands can be used to display specific sections of output. To enable the filtering command, enter a pipe (|) character after the **show** command and then enter a filtering parameter and a filtering expression.
- There are four filtering parameters that can be configured after the pipe:
 - section - Shows the entire section that starts with the filtering expression.
 - include - Includes all output lines that match the filtering expression.
 - exclude - Excludes all output lines that match the filtering expression.
 - begin - Shows all the output lines from a certain point, starting with the line that matches the filtering expression

Verify Directly Connected Networks

Command History Feature



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- The command history feature is useful because it temporarily stores the list of executed commands to be recalled.
 - To recall commands in the history buffer, press **Ctrl+P** or the **Up Arrow** key. The command output begins with the most recent command. Repeat the key sequence to recall successively older commands. To return to more recent commands in the history buffer, press **Ctrl+N** or the **Down Arrow** key. Repeat the key sequence to recall successively more recent commands.
 - By default, command history is enabled and the system captures the last 10 command lines in its history buffer. Use the **show history** privileged EXEC command to display the contents of the buffer.
 - It is also practical to increase the number of command lines that the history buffer records during the current terminal session only. Use the **terminal history size** user EXEC command to increase or decrease the size of the buffer.